

Research paper

Alterations of BDNF, mGluR5, Homer1a, p11 and excitatory/inhibitory balance in corticolimbic brain regions of suicide decedents

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ABSTRACT

Background: Developing biological based approaches for preventing suicide has become a priority. In recent years, there has been a surge in studies investigating the role of the glutamatergic system in suicide, although it remains unclear.

Methods: We evaluated changes in the gene expression of the metabotropic glutamate receptor 5 (mGluR5) and its scaffolding proteins Homer1a and p11 in the dorsolateral prefrontal cortex (DLPFC), amygdala (AMY), and hippocampus (HIP) of 28 suicide decedents (S) (with no clinical psychiatric history or treatment with anxiolytics or antidepressants) and 26 controls (C) by real-time PCR (qPCR). Indeed, we measured BDNF gene expression and VGluT1 and VGAT immunoreactivities in the HIP by qPCR and immunohistochemistry, respectively. Cases and controls matched for age (C: 48.6 ± 11.6 years; S: 46.9 ± 14.5 years) and postmortem interval (PMI; C: 20.1 ± 13 h; S: 16.9 ± 5 h).

Results: In DLPFC, S had lower p11 gene expression levels, but no differences were found in mGluR5 or Homer1a. In the AMY and HIP, mGluR5 and Homer1a were increased, p11 and BDNF were reduced. In the HIP, there were less VGAT-ir and more VGluT1-ir.

Limitations: Future studies are necessary to evaluate protein levels, and determine the cell types and potential compensatory mechanisms in a larger sample including S diagnosed with psychiatric disorders, females and different ethnicities.

Conclusions: This study identified significant alterations in mGluR5, Homer1a, p11, BDNF and excitatory/inhibitory balance in corticolimbic brain areas of S. These results further characterize the biological basis of suicide, contributing to the identification of potential biomarkers for suicide prevention.

1. Introduction

Suicide is a common and devastating health problem worldwide, accounting for almost 1.4 % of all deaths and the 17th leading cause of death. In most developed countries, suicide is the second leading cause of death among people aged 15–29 years and the first for men under 40 (WHO, 2018). Given the social and economic impact of suicide, preventing suicide has become an essential priority for modern societies. Accordingly, several efforts were carried out to set up effective and evidence-based interventions to prevent suicide (WHO, 2019). The

characterization of the neurobiological mechanisms underlying suicide is one of the significant challenges to identifying biomarkers for guiding prevention or becoming potential targets for the treatment of patients at risk.

Converging evidence pointed out the glutamatergic system as a potential target for reducing suicide risk. Glutamate concentrations significantly changed in the plasma and cerebrospinal fluid of depressive and suicide decedents (Frye et al., 2007; Levine et al., 2000). Recently, ketamine, an antagonist of NMDA receptors, reduced suicidal ideation (Ahmed et al., 2023; Chen et al., 2019; Domany et al., 2020; Hochschild

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et al., 2021; Kritzer et al., 2022; Lascelles et al., 2019; Lengvenyte et al., 2019; Witt et al., 2020). Interestingly, esketamine, the S-enantiomer of ketamine, was approved in USA and Europe as an intranasal formulation for treatment-resistant depression due to its immediate effect in reducing depressive symptoms and suicidal ideation (Bahr et al., 2019; Canuso et al., 2018; Daly et al., 2018). Both drugs increase glutamate concentrations by blocking NMDA receptors. Consequently, AMPA receptors are activated, and subsequent activation of mTORC1 and BDNF-TrkB signaling improves neurogenesis (Hashimoto, 2020; Molero et al., 2018; Scotton et al., 2023). Thus, the final increase in neurogenesis may underlie, at least in part, the efficacy of NMDA receptor antagonists in reducing depressive symptoms and suicide.

The function of NMDA receptors is regulated by the metabotropic glutamate receptor 5 (mGluR5) (Clifton et al., 2013; Homayoun et al., 2004; Uslaner et al., 2009), which is linked with mood disorders and suicide (Crupi et al., 2019; Davis et al., 2019; Esterlis et al., 2018b). The mGluR5 modulates neuronal plasticity by regulating K⁺ channels and glutamate release (Kato et al., 2012). Evidence achieved to date in animal models of depression revealed changes in mGluR5 depending on the animal model and the brain regions analyzed.

A reduction of the mGluR5 protein and density was described in several brain regions of the Flinders Sensitive Line of rats (Kovacevic et al., 2012). In rats exposed to the unpredictable chronic mild stress model, alterations of mGluR5 were found in different fields of the HIP, observing an increase in the CA1 and a decrease in the CA3 (Wieronska et al., 2001). Furthermore, mice lacking the mGluR5 showed depressive-like behaviors (Shin et al., 2015; Xu et al., 2021), and the blockade of the mGluR5 induced an antidepressant-like effect (Chaki et al., 2013; Lee et al., 2015a; Palucha-Poniewiera et al., 2013). Additional studies using mGluR5 partial/full negative allosteric modulators or inverse agonists supported the antidepressant-like effects of the pharmacological modulation of mGluR5 (for more detail (Dogra and Conn, 2021)).

In humans, studies evaluating alterations of mGluR5 in brain regions of psychiatric patients revealed opposite changes, including no modification, lower or higher expression (for more detail (Asch et al., 2023)). Focusing on suicide, upregulation of the mGluR5 gene and protein expressions was reported in crucial brain regions of the corticolimbic circuit of patients with major depressive disorder (MDD) (who died mainly by suicide) (Chandley et al., 2014), PTSD, and suicidal ideation (Davis et al., 2019). Interestingly, treatment with ketamine significantly down-regulated mGluR5 in MDD patients, and this action was associated with its efficacy as an antidepressant and reducing suicide risk (Esterlis et al., 2018a). Thus, modulation of mGluR5 has been proposed as a potential new strategy to achieve antidepressant effects (Domin et al., 2014; Wang et al., 2020; Zhang et al., 2017).

Based on these findings, scaffolding proteins regulating mGluR5 availability on the surface of neurons and astrocytes have become relevant (Piers et al., 2012). The glutamatergic scaffolding proteins, SHANK-1 and PSD-65, were increased in PTSD patients (Holmes et al., 2017) and suicide decedents (Dean et al., 2016). More recently, additional glutamatergic scaffolding proteins, Homer1 and p11, were associated with the regulation of emotional responses and the efficacy of antidepressants (Chen et al., 2022; Sun et al., 2021; Svenningsson et al., 2013; Veldman et al., 2021). Homer1 gene encodes different transcriptional variants, including the truncated forms Homer1a, which promotes the trafficking of the mGluR5 from the cell body to dendrites and axons in the hippocampus (HIP) (Ango et al., 2000; Ango et al., 2002), regulating synaptic architecture and, consequently, learning and memory. Genetic and pharmacological studies in rodents demonstrated that Homer1a is involved in emotional reactivity, anxiety, and depression (Igaz et al., 2004; Lominac et al., 2005; Tronson et al., 2010). In humans, single nucleotide polymorphisms of Homer1 were associated with MDD (Rietschel et al., 2010), mood state in schizophrenic patients (Norton et al., 2003; Spellmann et al., 2011), psychotic symptoms in Parkinson's disease (De Luca et al., 2009), the response to antidepressants (Rietschel et al., 2010), and antipsychotics (Spellmann et al., 2011). Additional

Homer1 gene variants were associated with suicidal behaviors (Rao et al., 2016; Strauss et al., 2012).

Likewise, animal studies indicated that the scaffolding protein p11 (S100a10), a member of the S100 family of proteins, is involved in the antidepressant-like effects of mGluR5-antagonists, increasing mGluR5 surface availability and signaling function (Lee et al., 2015a, 2015b). mGluR5 and p11 mutually facilitate their accumulation at the plasma membrane. Besides, p11 was associated with mood disorders and suicide (Anisman et al., 2008; Zhang et al., 2011) and, even more interestingly, with the efficacy of conventional and recent antidepressant drugs such as ketamine (Chen et al., 2022; Grossert et al., 2019; Sun et al., 2016a; Svenningsson et al., 2006; Svenningsson et al., 2013; Veldman et al., 2021; Wang et al., 2022).

Furthermore, chronic stress increases Homer1-mGluR5 signaling leading to a downregulation of the brain-derived neurotrophic factor (BDNF), a neuroprotective factor widely distributed in the brain that plays a relevant role in regulating synaptic plasticity and memory consolidation. Detriments in BDNF have been associated with increased vulnerability to stress and mood, including suicide (Notaras and van den Buuse, 2020; Paska et al., 2013; Phillips, 2017; Xu et al., 2016; Zai et al., 2012; Zaletel et al., 2017). mGluR5 antagonism induced antidepressant-like effects and modulated PTSD-like behaviors by increasing TrkB-BDNF (Cheng et al., 2023). Notably, the increase of p11 by a mechanism depending on BDNF-TrkB activity is also involved in the fast-antidepressant action of ketamine (Chottekalapanda et al., 2020; Warner-Schmidt et al., 2010).

Currently, the balance of glutamate/GABA appears to play a relevant role in the pathophysiology of different psychiatric disorders and its control in the mechanisms of action of antidepressants. Interestingly, mGluR5 modulates the activity of glutamatergic neurons and GABAergic interneurons (De Blasi et al., 2001), being the inhibition of GABAergic signaling and the consequent enhancement of glutamate activity the mechanism underlying the fast-antidepressant-like effects of mGluR5 antagonism in rodents (Lee et al., 2015a; Li et al., 2006). Moreover, the fast-antidepressant effects of ketamine are related also to an increase of glutamate activity by transient inhibition of inhibitory interneurons, in which p11 may play a role (Duman et al., 2019; Fogaca et al., 2021; Gerhard et al., 2020; Lazarevic et al., 2021).

Based on these emerging data, here, alterations of mGluR5, Homer1a, p11, BDNF and excitatory/inhibitory balance were evaluated in different corticolimbic brain regions of suicide decedents. We examined the relative gene expression of the mGluR5, Homer1a, and p11 in the DLPFC, AMY, and HIP of suicide decedents with no clinical psychiatric history or treatment with anxiolytics or antidepressants. Moreover, we measured BDNF gene expression and VGluT1 and VGAT immunoreactivities in the HIP by qPCR and immunohistochemistry, respectively. The DLPFC plays a crucial role in decision-making and cognitive functions. Dysfunctions in the DLPFC of patients who attempted suicide are associated with impaired executive functions and increased impulsivity (Jollant et al., 2005; van Heeringen et al., 2003). On the other hand, impulsive behaviors are closely related to greater susceptibility to suicide (Bridge et al., 2012; Jollant et al., 2007; Mann et al., 1999; van Heeringen et al., 2011). The AMY is involved in stress, threat and negative affect; importantly, changes in its functional connectivity appears to be related with suicide ideation (Alarcon et al., 2019; Ballard et al., 2020; Wang et al., 2019; Wei et al., 2018). Finally, the HIP is a vulnerable brain region to neuropsychiatric damage, being part of the limbic-cortical-hypothalamic circuit, which is implicated in the pathophysiology of suicide (Lutz et al., 2017; Mahar et al., 2017). Alterations in the HIP, including anatomical and histopathological changes, have been identified in psychiatric patients who died by suicide or with suicide ideation (Chen et al., 2020; Zhang et al., 2022).

The nature of these unique samples, medication free S, ensures that the alterations observed are not due to chronic pharmacological treatment with antidepressants or other drugs usually employed in these patients.

2. Materials and methods

2.1. Subjects

Brain samples were obtained from autopsy material provided by the Institute of Legal Medicine of Alicante following the agreement signed between the Miguel Hernandez University of Elche, the Institute of Legal Medicine of Alicante, and the Department of Justice of the “Comunidad Valenciana” (all located in Spain) for analyzing brain samples from autopsies.

Table 1 presents demographic and postmortem data for included subjects, who were all white and came from the Spanish region of Alicante. Dorsolateral Prefrontal Cortex (DLPFC) (Brodmann area 9), amygdaloid nucleus (AMY), and hippocampus (HIP) were dissected from suicide decedents (S, $n = 28$; age range, 18–78) and controls (C, $n = 26$; age range, 19–65). All cases included in the present study were male and were matched as closely as possible for age (C: 48.4 ± 11.6 years; S: 46.9 ± 14.5 years) and postmortem interval (PMI, C: 20.1 ± 13 h; S: 16.9 ± 5 h). An additional postmortem interval consisting of the time from the death until the introduction of the corpse in a refrigerated chamber (4°C) was considered (PMR, C: 3.3 ± 2.8 h; S: 4 ± 2.6 h). Individuals with a history of cerebral trauma, central nervous system disease, chronic alcoholism, or illicit or therapeutic drug use were excluded. Body fluids (blood, bile, aqueous humor, and urine) were used for toxicological screening of cocaine, opiates, alcohol, antidepressants, and other acidic or basic drugs, being a positive result an exclusion criterion. No control or suicide subjects meet the criteria for any psychiatric disorder diagnosis during their lifetime, according to close relatives, forensic psychological assessment, and clinical history.

Moreover, there were no previous self-harm, suicide attempts, or relatives' history of suicidality reported by interviews with family members. All C and S were Caucasian from Spain (Alicante region). The brain samples were coded to mask the investigators the cause of death. The local ethics committee approved the collection of tissues.

2.2. Brain dissection

Brains were obtained 17–20h after death (17 ± 5 h and 20 ± 13 h for S and C, respectively). After removal from the skull, brains were dissected into six major pieces (four cortical lobes, basal ganglia–diencephalon, and lower brainstem–cerebellum), according to Brodmann's Atlas. Immediately, a fine Graefe's knife carefully dissected regions of interest (DLPFC (Brodmann area 9), AMY, and HIP) on a glass surface with dry ice. The white matter was removed, frozen on dry ice, and stored at -80°C until further use.

2.3. RNA integrity number (RIN) evaluation

Total RNA was isolated from DLPFC, AMY, and HIP snap-frozen tissue using TRI Reagent (Applied Biosystems, Foster City, CA). Subsequently, we used the 2100 Bioanalyzer (Agilent Technologies, Madrid, Spain) to analyze the integrity of RNA samples by detecting RNA ribosomal bands (18S and 28S). Depending on the ratio 28S/18S, we calculated a value between 1 and 10, called the RNA integrity number (RIN), as a measure of RNA degradation (with lower values expressing more significant degradation). All RNA samples from C and S presented a mean RIN value higher than 6 (DLPFC: C = 6.26 ± 0.39 , range = 5.66–6.86; S = 6.4 ± 0.32 ; range = 5.78–6.82) (AMY: C = 6.18 ± 0.4 , range = 5.45–6.65; S = 6.5 ± 0.2 ; range = 5.90–6.93) (HIP C = 6.8 ± 0.29 , range = 6.19–7.39; S = 6.2 ± 0.42 ; range = 5.95–6.99), which represents good RNA quality for gene expression analyses (Fleige and Pfaffl, 2006; Fleige et al., 2006; Perez-Ortiz et al., 2013).

2.4. Real time qPCR analysis of mGluR5, Homer1a, and p11

Relative gene expression analyses of the mGluR5, Homer1a, and p11 in DLPFC, AMY, and HIP were carried out in C, and S. BDNF was measured in the HIP of C, and S. Total RNA was obtained from brain sections using Tri Reagent extraction reagent (Applied Biosystems, Madrid, Spain). After DNase digestion, reverse transcription followed the manufacturer's instructions (Applied Biosystems, Madrid, Spain).

Table 1
Demographic and postmortem characteristics of suicidal and control samples.

Sample	Age	PMI (h)	PMR (h)	Cause of death	Sample	Age	PMI (h)	PMR (h)	Cause of death
C1	49	7	4	Natural, Cardiac arrest	S1	45	11	9h30'	Suicide, Hanging
C2	47	22	1h30'	Natural, Acute respiratory failure	S2	61	23	1h30'	Suicide, Hanging
C3	52	23	3h30'	Natural, Cardiac arrest	S3	52	18	3h30'	Suicide, Hanging
C4	64	16	6	Natural, Sudden death	S4	57	20	8	Suicide, Hanging
C5	56	24	2h30'	Natural, Cardiac arrest	S5	25	7	3	Suicide, Hanging
C6	55	6	2	Natural, Acute pancreatitis	S6	40	17	2h30'	Suicide, Hanging
C7	33	24	1h30'	Natural, Sudden death	S7	41	10	2h30'	Suicide, Hanging
C8	52	10	7.5	Natural, Myocardial infarction	S8	42	8	1h30'	Suicide, Hanging
C9	40	19	1h30'	Natural, Pulmonary embolism	S9	30	20	4	Suicide, Hanging
C10	64	18	2h30'	Natural, Cardiac hypertrophy	S10	36	21	5	Suicide, Hanging
C11	19	6	4h30'	Natural, Sudden death	S11	41	18	8	Suicide, Hanging
C12	40	19	2	Natural, Sudden death	S12	22	18	6	Suicide, Hanging
C13	38	11	1h30'	Natural, Sudden death	S13	66	27	5	Suicide, Hanging
C14	50	20	7h30'	Natural, Sudden death	S14	18	12	2	Suicide, Jumping from a height
C15	39	4	3	Natural, Sudden death	S15	66	18	2h30'	Suicide, Hanging
C16	51	20	9	Natural, acute respiratory failure	S16	78	20	2h30'	Suicide, Hanging
C17	47	16	1	Natural, Cardiac hypertrophy	S17	46	21	2h30'	Suicide, Hanging
C18	52	36	1	Natural, Rupture of aortic aneurism	S18	36	18	2h30'	Suicide, Hanging
C19	65	12	–	Natural, Heart failure	S19	58	16	0.5	Suicide, Jumping from a height
C20	37	7	3	Natural, Heart failure	S20	42	12	3	Suicide, Jumping from a height
C21	64	52	12	Natural, Cardiac arrest	S21	44	19	9	Suicide, Hanging
C22	35	41	0.25	Natural, Suffocation	S22	70	20	4h30'	Suicide, Hanging
C23	38	48	1h30'	Natural, Sudden death	S23	44	20	7	Suicide, Hanging
C24	60	23	1	Natural, Myocardial infarction	S24	50	16	0.5	Suicide, Jumping from a height
C25	63	–	–	Natural, Myocardial infarction	S25	56	10	–	Suicide, Jumping from a height
C26	54	–	2	Natural, Sudden death	S26	45	22	3	Suicide, Hanging
					S27	41	12	3	Suicide, Jumping from a height
					S28	61	19	9	Suicide, Hanging

C: normal control; S: suicide; PMI: postmortem interval; PMR: postmortem interval until refrigeration (4°C).

Relative mGluR5, Homer1a, p11, and BDNF gene expression levels were measured using TaqMan Gene Expression Assays (Hs00168275_m1, Hs01029333_m1, Hs00751478_s1, and Hs02718934_s1 respectively) as a double-stranded DNA-specific fluorescent dye and performed on the Step One Plus Sequence Detector System (Applied Biosystems, Madrid, Spain). In this study, two housekeeping genes, cyclophilin (PPIA, Hs99999904_m1) and synaptophysin (SYP, Hs00300531_m1), were used to ensure the validity and reproducibility of results. All primer-probe combinations were optimized and validated for relative quantification of gene expression. Briefly, samples were run in duplicates for each target and housekeeping gene. Sample values were considered when standard deviations between duplicates were less than ± 0.3 to avoid inhomogeneous pipetting. Data for each target gene were normalized to the endogenous reference genes. The fold change in target gene abundance was determined using the $2^{-\Delta\Delta Ct}$ method (Livak and Schmittgen, 2001) so that S levels were expressed relative to C levels. For each sample, we represent the results as the mean obtained with each endogenous housekeeping gene (PPIA and SYP).

Immunohistochemistry

The general criteria reported by Amaral and Lavenex (2007) were used to define the hippocampal areas and strata. The HIP comprises the DG, the HIP proper or CA (containing CA₄, CA₃, CA₂, and CA₁), the subiculum, and the entorhinal cortex. In this study, we focused on the DG field due to its role in learning and memory. Proper synaptic plasticity and excitatory-inhibitory balance in the DG are essential to the efficient processing of the information (Jedlicka et al., 2018). Abnormalities in plasticity DG has been observed in different psychiatric disorders and suicide (Boldrini et al., 2013; Cobb et al., 2013; Hanson et al., 2015; Huang et al., 2013; Wang et al., 2010).

Briefly, six coronal sections of the HIP at the level of the hippocampal body were cut with a Microm HM 650v vibratome (Thermo Fisher Scientific, Inc., Barcelona, Spain) at 100 μ m and stored in 0.05 % sodium azide in PB at 4 °C. The slices were double immunostained for fluorescence, starting with guinea pig anti-vesicular glutamate transporter type 1 (VGluT1) antibody (1:1000; Synaptic Systems) and then rabbit anti-vesicular GABA transporter (VGAT) antibody (1:1000; Synaptic Systems). All sections were then incubated with goat anti-guinea pig antibody, Alexa Fluor 488 labeled (1:200, Molecular Probes, Invitrogen, Barcelona, Spain), followed with goat biotinylated anti-rabbit antibody Alexa Fluor 594 labeled (1:200, Molecular Probes, Invitrogen, Barcelona, Spain). Sections were mounted using ProLong Gold (Molecular Probes, Invitrogen). All samples were examined in a Leica SPEII confocal laser fluorescence microscope, with images captured using Leica Application Suite X software.

2.5. Statistical analyses

Statistical analyses were performed using the Student's *t*-test to compare S and C groups. Pearson product-moment correlation was used to rule out potential correlations between RIN, PMR, and PMI on the one hand and the results of gene expression studies on the other. We used SigmaPlot software (Systat Software Inc., Chicago, IL, USA) for all analyses and considered differences to be significant if the error probability was $< 5\%$. P-values were corrected for multiple comparisons using the False Discovery Rate method.

3. Results

3.1. Relative mGluR5, Homer1a, and p11 gene expressions in the DLPFC of suicide decedents

As shown in Fig. 1, there were no significant differences in mGluR5 (Fig. 1A, Student's *t*-test; $t = -1.351$, $p = 0.203$, 50 df) nor in Homer1a gene expressions (Fig. 1B, Student's *t*-test; $t = -0.757$, $p = 0.453$, 46 df) between S and C in the DLPFC. Interestingly, S presented a significant reduction of p11 in this brain region (-39%) (Fig. 1C, Student's *t*-test; $t = 3.097$, $p = 0.006$, 45 df).

Relative mGluR5, Homer1a, and p11 gene expressions in the AMY of suicide decedents.

Quantitative PCR studies showed significantly higher gene expression of mGluR5 ($+29\%$) (Fig. 2A, Student's *t*-test; $t = -4.836$, $p < 0.0022$, 51 df) and Homer1a ($+23\%$) in the AMY (Fig. 2B, Student's *t*-test; $t = -2.58$, $p = 0.018$, 46 df) of S compared with C. In contrast, a significant reduction in p11 was found in S compared with C (-30%) (Fig. 2C, Student's *t*-test; $t = 2.407$, $p = 0.026$, 41 df).

3.2. Relative mGluR5, Homer1a, p11, and BDNF gene expressions in the HIP of suicide decedents

Statistical analyses showed that S expressed higher mGluR5 ($+30\%$) (Fig. 3A, Student's *t*-test; $t = -3.893$, $p < 0.0022$, 51 df) and Homer1a gene expressions in the HIP ($+29\%$) (Fig. 3B, Student's *t*-test; $t = -3.026$, $p = 0.006$, 41 df). Interestingly, a significant reduction of the relative p11 (-48%) (Fig. 3C, Student's *t*-test; $t = 6.139$, $p < 0.0022$, 44 df) and BDNF gene expressions (-26%) (Fig. 3D, Student's *t*-test; $t = 5.344$, $p < 0.0022$, 44 df) were found in the HIP of S in comparison to C.

In addition, a Pearson product-moment correlation was applied to calculate a correlation coefficient between RIN, PMR, and PMI and between gene expression values for mGluR5, Homer1a, p11, and BDNF of C and S cases. There was no correlation between any of these parameters and gene expression values. Furthermore, we also analyzed the correlation coefficient between age and gene expression values for mGluR5,

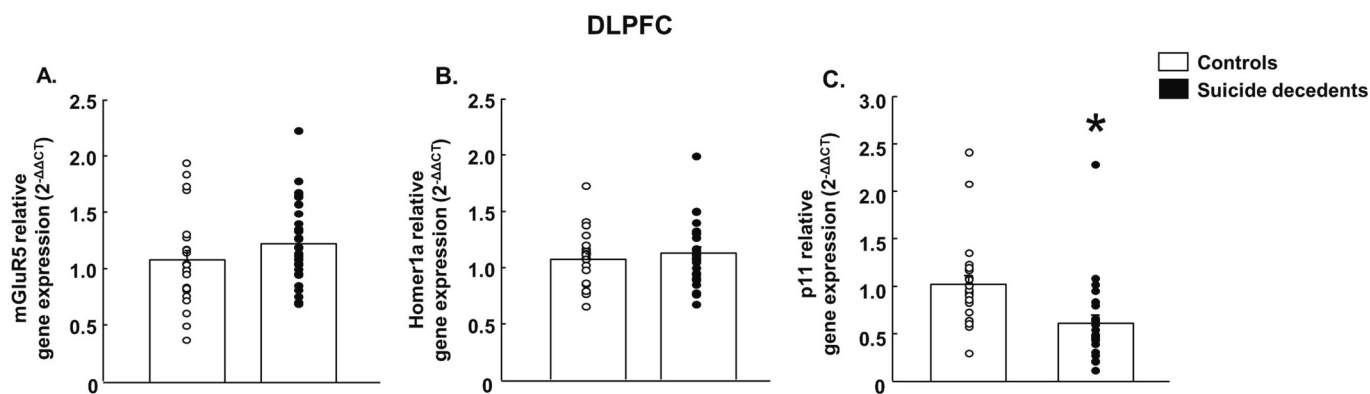


Fig. 1. Relative mGluR5 (A), Homer1a (B) and p11 (C) gene expression in the dorsolateral prefrontal cortex of suicide decedents and controls analyzed by real time PCR. Gene expression in suicide decedents is expressed relative to control subjects. The results are expressed as mean $\pm$ SEM of $2^{-\Delta\Delta Ct}$. * Values from suicide decedents significantly differ ($p < 0.05$) from control subjects. DLPFC: dorsolateral prefrontal cortex.

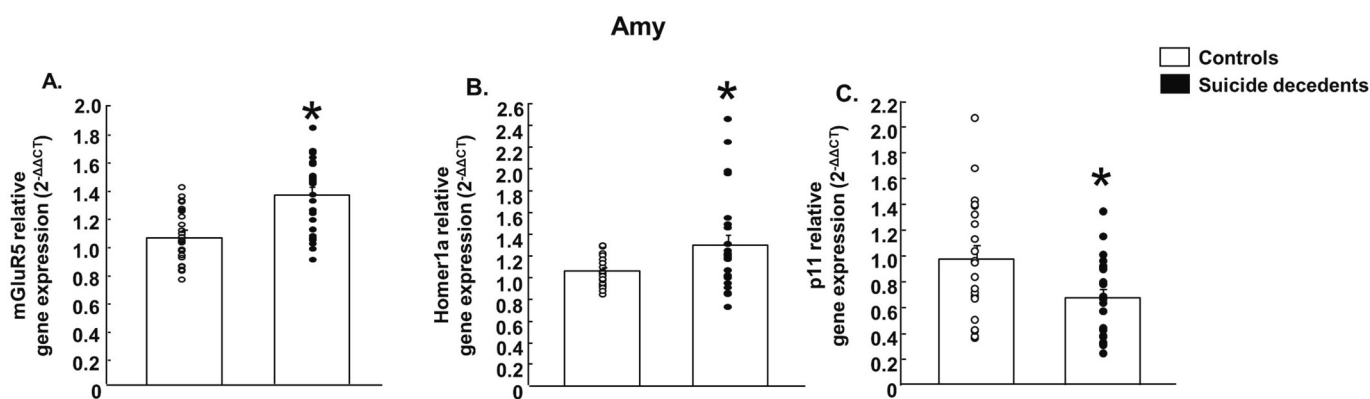


Fig. 2. Relative mGluR5 (A), Homer1a (B) and p11 (C) gene expression in the amygdala of suicide decedents and controls analyzed by real time PCR. Gene expression in suicide decedents is expressed relative to control subjects. The results are expressed as mean $\pm$ SEM of $2^{-\Delta\Delta Ct}$. * Values from suicide decedents significantly differ ($p < 0.05$) from control subjects. Amy: amygdala.

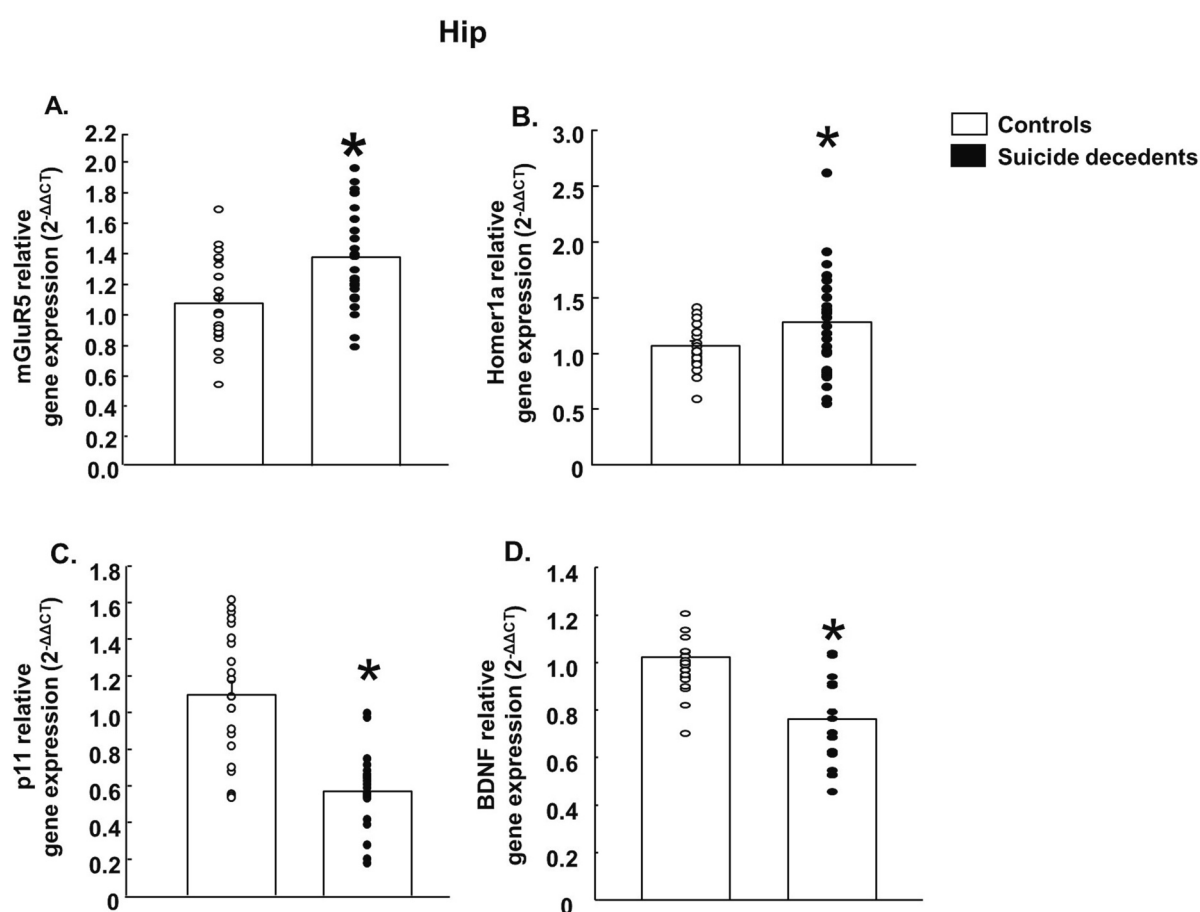


Fig. 3. Relative mGluR5 (A), Homer1a (B), p11 (C) and BDNF (D) gene expression in the hippocampus of suicide decedents and controls analyzed by real time PCR. Gene expression in suicide decedents is expressed relative to control subjects. The results are expressed as mean $\pm$ SEM of $2^{-\Delta\Delta Ct}$. * Values from suicide decedents significantly differ ($p < 0.05$) from control subjects. Hip: hippocampus.

Homer1a, p11, and BDNF in C and S cases, finding no significant association (Table 2).

3.3. Immunolabeling of VGluT1 and VGAT in DG of the HIP

The confocal photograph showed that the distribution of both VGluT1-ir and VGAT-ir boutons in DG in S was abnormal compared with C and revealed an alteration of the laminar distribution in excitatory and inhibitory input. In the DG, the most prominent finding was a decrease

in the VGAT-ir boutons in the granular layer (Fig. 4A, B, C, D). The areas occupied by the VGAT-ir boutons were the largest in the granular layer in the S group. In addition, a very high VGluT1-ir boutons in the hilus of the DG was observed in S compared with C (Fig. 4A, B, E, F).

4. Discussion

The results of this study identify dysregulations in key elements of the glutamatergic system and neuroplasticity in corticolimbic brain

Table 2

Pearson coefficient correlation for mGluR5, Homer1a and p11 gene expressions and RIN, PMI, PMR and age variables for control and suicide cases.

			RIN*	PMI*	PMR*	Age
PFC	Control cases	mGluR5	r =	r =	r =	r = 0.133
			−0.0374	0.106	−0.370	p = 0.544
			p = 0.872	p =	p =	
				0.649	0.108	
		Homer1a	r =	r =	r =	r =
			−0.109	0.0783	0.335	0.0515
			p = 0.629	p =	p =	p = 0.820
				0.736	0.149	
	Suicide cases	mGluR5	r =	r =	r =	r = 0.297
			−0.181	0.0877	−0.0569	p = 0.118
			p = 0.387	p =	p =	
				0.651	0.787	
		Homer1a	r =	r =	r =	r =
			−0.222	0.0616	−0.321	−0.195
			p = 0.321	p =	p =	p = 0.373
AMY	Control cases	mGluR5	r =	r =	r =	r =
			0.0478	−0.263	−0.326	−0.298
			p = 0.837	p =	p =	p = 0.158
				0.249	0.161	
		Homer1a	r =	r =	r =	r =
			−0.0863	0.277	0.0352	−0.00878
			p = 0.734	p =	p =	p = 0.972
	Suicide cases	mGluR5	r =	r =	r =	r =
			−0.322	0.110	0.0727	−0.141
			p = 0.117	p =	p =	p = 0.476
				0.570	0.736	
		Homer1a	r = 0.371	r =	r =	r =
			p = 0.173	0.0289	−0.173	−0.221
HIP	Control cases	mGluR5	r =	r =	r =	r =
			−0.255	0.144	0.280	0.0976
			p = 0.265	p =	p =	p = 0.650
				0.534	0.233	
		Homer1a	r =	r =	r =	r =
			−0.0521	0.127	0.165	0.0676
			p = 0.866	p =	p =	p = 0.818
	Suicide cases	mGluR5	r =	r =	r =	r = 0.139
			−0.0279	−0.215	0.203	p = 0.491
			p = 0.894	p =	p =	
				0.282	0.341	
		Homer1a	r = 0.185	r =	r =	r = 0.189
			p = 0.421	−0.120	−0.231	p = 0.376
HIP	Control cases	mGluR5	r =	r =	r =	r =
			−0.255	0.144	0.280	0.0976
			p = 0.265	p =	p =	p = 0.650
				0.534	0.233	
		Homer1a	r =	r =	r =	r =
			−0.0521	0.127	0.165	0.0676
			p = 0.866	p =	p =	p = 0.818
	Suicide cases	mGluR5	r =	r =	r =	r = 0.139
			−0.0279	−0.215	0.203	p = 0.491
			p = 0.894	p =	p =	
				0.282	0.341	
		Homer1a	r = 0.185	r =	r =	r = 0.189
			p = 0.421	−0.120	−0.231	p = 0.376
HIP	Control cases	mGluR5	r =	r =	r =	r =
			−0.255	0.144	0.280	0.0976
			p = 0.265	p =	p =	p = 0.650
				0.534	0.233	
		Homer1a	r =	r =	r =	r =
			−0.0521	0.127	0.165	0.0676
			p = 0.866	p =	p =	p = 0.818
	Suicide cases	mGluR5	r =	r =	r =	r = 0.139
			−0.0279	−0.215	0.203	p = 0.491
			p = 0.894	p =	p =	
				0.282	0.341	
		Homer1a	r = 0.185	r =	r =	r = 0.189
			p = 0.421	−0.120	−0.231	p = 0.376
HIP	Control cases	mGluR5	r =	r =	r =	r =
			−0.255	0.144	0.280	0.0976
			p = 0.265	p =	p =	p = 0.650
				0.534	0.233	
		Homer1a	r =	r =	r =	r =
			−0.0521	0.127	0.165	0.0676
			p = 0.866	p =	p =	p = 0.818
	Suicide cases	mGluR5	r =	r =	r =	r = 0.139
			−0.0279	−0.215	0.203	p = 0.491
			p = 0.894	p =	p =	
				0.282	0.341	
		Homer1a	r = 0.185	r =	r =	r = 0.189
			p = 0.421	−0.120	−0.231	p = 0.376
HIP	Control cases	mGluR5	r =	r =	r =	r =
			−0.255	0.144	0.280	0.0976
			p = 0.265	p =	p =	p = 0.650
				0.534	0.233	
		Homer1a	r =	r =	r =	r =
			−0.0521	0.127	0.165	0.0676
			p = 0.866	p =	p =	p = 0.818
	Suicide cases	mGluR5	r =	r =	r =	r = 0.139
			−0.0279	−0.215	0.203	p = 0.491
			p = 0.894	p =	p =	
				0.282	0.341	
		Homer1a	r = 0.185	r =	r =	r = 0.189
			p = 0.421	−0.120	−0.231	p = 0.376

Table 2 (continued)

	RIN*	PMI*	PMR*	Age
BDNF	r =	r =	r =	r = 0.188
	−0.0359	0.229	−0.191	p = 0.440
	p = 0.830	p =	p =	
		0.360	0.447	

(*) RIN: RNA integrity number; PMI: postmortem interval; PMR postmortem interval until refrigeration (4 °C).

regions (DLPFC, AMY, and HIP) of medication free suicide decedents never diagnosed with mental illness at the time of death. The following observations support this assumption: 1) mGluR5 gene expression and its scaffolding protein Homer1a was increased in the AMY and HIP of S without no alterations in the DLPFC, 2) The gene expression of the scaffolding protein P11 was reduced in all the brain regions analyzed and 3) A reduction of BDNF gene expression and alterations of excitatory/inhibitory balance were found in the HIP of S compared with C.

Approximately 90 % of suicide cases are patients with a psychiatric or drug-use disorder; however, 10 % are people with no diagnosis of mental illness. Most postmortem studies carried out on psychiatric-suicide decedents, mainly in depressive and schizophrenic patients, are involved in the identification of crucial targets underlying suicidal behavior. Pharmacological management of these disorders is complex, and patients are treated with anxiolytic, antidepressant and/or antipsychotic drugs. The effects of long-term drug treatment may distort the results in these postmortem studies. It is challenging to elucidate if the findings may result from the drug treatment. Access to brain samples of medication free suicide decedents never diagnosed with mental illness at the time of death is an excellent opportunity to analyze the molecular alterations of suicidal behavior *per se*, avoiding other potential confounding variables.

This study revealed increased mGluR5 gene expression in AMY and HIP of suicide decedents. These changes agree with previous studies showing increased mGluR5 gene expression in the *locus coeruleus* of depressive suicide patients (Chandley et al., 2014). Besides, abnormalities in mGluR5 availability in frontal and limbic areas were also identified in suicide PTSD patients (Davis et al., 2019). In our study, changes in mGluR5 gene expression are accompanied by similar alterations in Homer1a gene expression. This is the first study in which Homer1a gene expression was measured in different brain regions of suicide decedents, revealing significant changes in AMY and HIP. These findings support previous studies reporting common and genetic variants of the Homer1 gene associated with suicide ideation in Chinese and Caucasian populations (Rao et al., 2016; Rao et al., 2017; Rao et al., 2018; Strauss et al., 2012).

In contrast, no alterations were found in mGluR5 or Homer1a gene expression in the DLPFC. No previous *postmortem* studies analyzed the gene expression of both targets in the DLPFC. Only one study reported an increase of mGluR5 in DLPFC, ventromedial PFC, and orbitofrontal cortex (OFC) of suicide-PTSD but no alterations in suicide-MDD (Davis et al., 2019). Curiously, the inability to increase the expression of Homer1a in PFC was associated with increased vulnerability to chronic stress or impairments in memories of stressful events in rodents (Igaz et al., 2004; Lominac et al., 2005; Szumlinski et al., 2005; Vazdarjanova et al., 2002). Conversely, the overexpression of Homer1a in the HIP and the subsequent modulation of mGluR5 was associated with higher vulnerability to chronic stress (Wagner et al., 2015). Additional animal studies revealed that hippocampal Homer1 is modified differently in stress-resilient compared to -vulnerable animals (Reshetnikov and Bondar, 2021; Schmidt et al., 2010). Therefore, the pattern of expression of Homer1a and mGluR5 found in suicide decedents in these cortico-limbic areas (DLPFC, AMY, and HIP) may be related to stress response alterations in suicide.

The gene expression of p11 was significantly less in all the cortico-limbic regions analyzed of S compared to C. Previous studies identified a

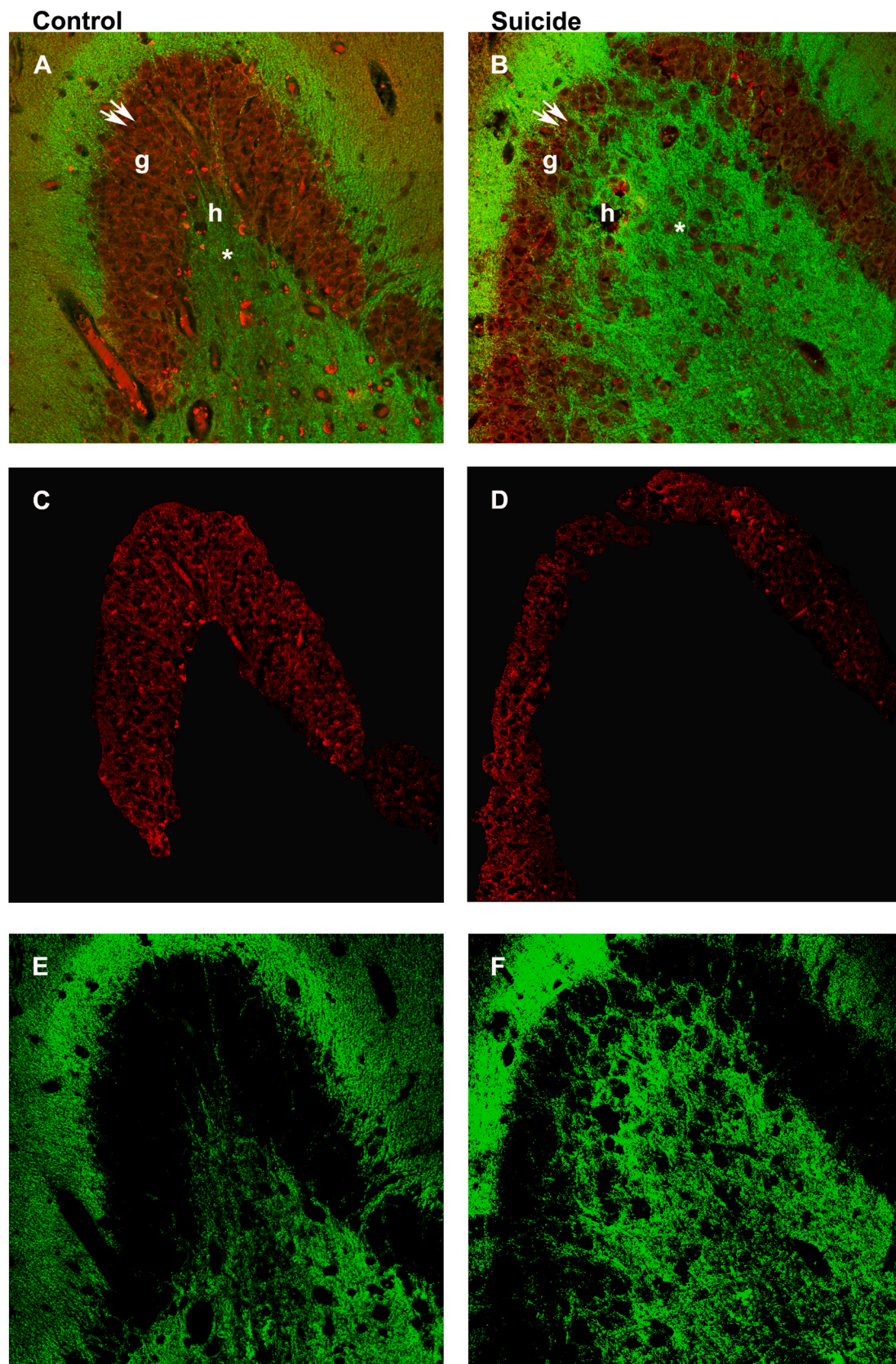


Fig. 4. Confocal immunolabeling of VGLut1 and VGAT in the dentate gyrus (DG) of suicide decedents and controls. Confocal photomicrographs showing VGLut1-ir (green labeling; E, F), VGAT-ir (red labeling; C, D), and merged images (A, B) in DG. Note the increased density of VGLut1-ir boutons in the hilus (asterisks in B and F) in the S group compared with the C group (asterisks in A and E). A decrease in VGAT-ir boutons density in the granular layer of DG was observed in the S group (double arrow in B and D) compared with C (double arrow in A and C) group. Same scale for (A–F). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

reduction of p11 gene expression in brain regions closely related to stress responses and mood state regulation, such as PFC, OFC, HIP, and AMY in suicide (Anisman et al., 2008; Zhang et al., 2011) and peripheral blood mononuclear cells (PBMCs) of suicide attempters (Zhang et al., 2011). In addition, significant reductions of p11 gene expression were found in the PFC of depressive and PTSD patients, both at high risk of suicide (Svenningsson et al., 2006; Zhang et al., 2008). Interestingly, the increase of p11 has been related to the efficacy of antidepressants (Chen et al., 2022; Svenningsson et al., 2006; Svenningsson et al., 2013). All the cumulative evidence achieved to date has pointed out p11 as a potential biomarker for depression and antidepressant response. Our results further support the involvement of p11 in emotional reactivity, suggesting that p11 may also be useful as a biomarker for preventing suicide.

The alterations in mGluR5, Homer1a and p11 were accompanied by a reduction of BDNF gene expression in the HIP of S compared to C. Previous studies revealed also a reduction of BDNF, RNA and protein levels, in the prefrontal cortex and HIP of suicide decedents (Dwivedi et al., 2003; Karege et al., 2005). More importantly, these findings agree with previous studies showing a connection between the glutamatergic system and neuroplasticity in mood disorders and suicide (Grady and Stahl, 2013; Myers et al., 2014; Popoli et al., 2011; Ryan et al., 2009). mGluR5 expression is often accompanied by changes in BDNF. Notably, a recent study showed that pharmacological inhibition of mGluR5 modifies BDNF/TrkB expression in the PFC and HIP, ameliorating PTSD-like behaviors in rats (Cheng et al., 2023).

The mGluR5 are interconnected with AMPA receptors. On the one hand, activation of mGluR5 modified AMPA signaling. On the other hand, AMPA receptors change mGluR5 signaling and facilitate downstream effects, including mTOR pathways and BDNF activation (Derkach et al., 2007; Henley and Wilkinson, 2016; Shepherd and Huganir, 2007). These mechanisms have been related to the efficacy of ketamine and esketamine in reducing suicide ideation (Koike and Chaki, 2014; Koike et al., 2011; Zhou et al., 2014).

In this context, Homer1a and p11 became relevant because both regulate, in a different way, mGluR5 signaling and plasticity (Ango et al., 2001; Ango et al., 2002; Kammermeier and Worley, 2007; Lee et al., 2015a, 2015b). In fact, Homer1a/BDNF/p11 signaling pathways have been associated with neuroprotection in an animal model of depression (Zuo et al., 2020). p11 is essential for the neuroprotective effects of BDNF (Sun et al., 2016b; Warner-Schmidt et al., 2010). Curiously, BDNF and ketamine promote the accumulation of Homer1a at synapses (Kato et al., 2003; Mahan et al., 2012; Serchov et al., 2016). It has been proposed that Homer1a may be the final common pathway at which elevated BDNF and p11 finally converge. Thus, mGluR5, p11, and Homer1a greatly impact neuroplasticity (Chen et al., 2022). Although future studies are needed to explore further the mechanisms by which these changes occur in suicide, it is tempting to speculate that alterations of Homer1a/mGluR5/p11 in limbic brain areas may be an important mechanism underlying the synaptic plasticity deficits observed in suicidal decedents (Drevets et al., 2008; Johnston et al., 2017; Soloff et al., 2012; Vai et al., 2020).

To further explore the role of glutamate neurotransmission in suicide, we analyzed potential alterations in the excitatory/inhibitory balance in the HIP. The balance of excitatory and inhibitory inputs of the neocortex and of the hippocampal formation, which includes the DG, and the CA is impaired in most psychiatric disorders such as autism spectrum disorders (ASD), schizophrenia, and MDD. Several studies suggest that MDD has associated with deficits in GABA inhibitory neurotransmission, including deficiencies in GABA_A and GABA_B (Levinson et al., 2010) and reduced expression of somatostatin between other cortical GABAergic interneurons (Tripp et al., 2011). Also, an association was reported between an altered glutaminergic and GABAergic balance and suicide risk. Some studies found a decrease in GABA_A subunits expression in suicide samples. Moreover, VGLUT expression was significantly higher in suicidal women (Powers et al., 2020). Here,

we observed a decrease in the VGAT boutons density in the granular layer and an increase in VGLUT1 density in the hilus of the HIP of S. In short, it was observed an alteration in excitatory and inhibitory balance in the DG of the HIP of suicide decedents.

In summary, this study reveals significant alterations in the glutamatergic system, neuroplasticity and excitatory/inhibitory balance in corticolimbic brain areas of suicide decedents. These findings further characterized the neuromolecular mechanisms underlying suicide. Complementary studies are necessary to examine if these targets' pharmacological modulation could help reduce suicide rates and to assess their clinical utility as biomarkers for preventing suicide.

4.1. Limitations

Despite S have no psychiatric diagnosis, it cannot be ruled out that they had undiagnosed psychiatric pathology. Moreover, all S and C are males and white. Additional large-scale studies will be necessary to evaluate if these alterations are reproducible in suicide decedents diagnosed with different psychiatric disorders, females and different ethnicities. Interestingly, access to PBMCs of patients with suicidal ideation and suicide attempts will support the potential of these targets as biomarkers for guiding suicide prevention.

An additional limitation of this study is that only analyzes changes in the gene expression of key targets of the glutamatergic system. Further studies are needed to examine if these changes are reflected at the protein level. Moreover, genome-wide and single-cell approaches should give information about compensatory mechanisms and whether these changes occur in specific cell types.

5. Conclusions

Our findings suggest an association between mGluR5/Homer1a/BDNF/p11 and excitatory/inhibitory balance alterations and suicide, although future studies should determine the specific contribution of these targets to suicidal behavior.

This study might be a major step for improving prevention and pharmacological management of suicide.

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CRediT authorship contribution statement

JM and MSGG were responsible for the study concept and design. MSGG, ABT and AV performed gene expression analyses, assisted with the data and interpretation of findings. MSGG and DN drafted the manuscript. JM revised the manuscript. All authors critically reviewed the content and approved the final version for publication.

Declaration of competing interest

The authors declare no conflict of interest.

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